

BOEKBESPREKINGS/BOOK REVIEWS

Success Epochs in Bernoulli Trials (with applications in number Theory)

by W. Vervaat

First Edition, 166 pages, Mathematical Centre Tracts 42, Mathematisch Centrum Amsterdam. Amsterdam, 1972.

Associated with a sequence of independent Bernoulli trials with probability p_k of success on the k -th trial, $k = 1, 2, \dots$, is a sequence of random variables L_n , $n = 1, 2, \dots$, which take as value the number of the trial on which the n -th success occurs. This paperback book derives limiting distributions of normalized partial sums defined in terms of the L_n , $n = 1, 2, \dots$, and the sequence of probabilities p_k , $k = 1, 2, \dots$. These theorems are applied to deriving probabilistic theorems on number representations on the interval $[0, 1]$. The class of number representations is due to Balkema (1968) and is a generalization of decimal expansions.

The most interesting theorem is that $n^{-1/2} \left(\sum_{k=1}^{L[nt]} p_k - nt \right)$ converges in distribution to the Wiener process if $p_k \rightarrow 0$ and $\sum_{k=1}^{\infty} p_k = \infty$. The most interesting technique employed is embedding sequences of independent Bernoulli trials in a Poisson process. This technique enables the author to relate the sequence of Bernoulli variables to an approximation of the Poisson process in which it is embedded and to apply properties of the Poisson process to derive limiting distributions.

The reviewer is a mathematical statistician and cannot judge the value of this book other than for its interest to a statistician. The limiting distributions do not appear to have any statistical applications. If the $p_k = 1/k$, the sequence of Bernoulli variables may be related to the sequence of successive maxima in independent random sampling. A statistical application of the limit theorems in the book may develop based on this fact. Other than this possibility, the reviewer does not see opportunities to apply the theory to statistics.

The theorems on numerical representations would not be of general interest to probabilists. This book is of interest only to a person already working in the area it covers.

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Statistical Thinking

by John L. Phillips, Jr.

First edition, XV + 124 pages, 56 Illustrations, 16 tables, W.H. Freeman and Company, San Francisco, 1973, £2.80 clothbound, £1.10 paperbound.

In the words of the author: "This book is not intended to be the main text of a statistics course. In such a course it should be assigned before work on the main text begins. Used in that way, it could be immensely helpful in breaking the psychological barrier that stands between many students and statistics — especially if statistics is a required course for those students. The most frequent function of the book, however, will probably be as basic reading in the statistics or measurement sections of courses that have broader content domains — courses in the behavioral sciences or in professional curricula (business, education, social work, and so forth) that are related to the social sciences."

I think the purpose of the book is met to a great extent. The essential formulae and inference techniques are given without any mathematical derivations. A few very interesting and well chosen examples and figures that explain the basic concepts are included. Descriptive statistics is dealt with in chapters two to four while chapter five is devoted to correlation. It is a pity that only ungrouped data are treated in these chapters and I think this is a serious shortcoming. On the other hand chapter five includes the concepts of rank correlation and the difference between correlation and causality is explained well. In chapters six and seven a little background (normal distribution and distribution of sample mean, etc.) for statistical inference is given and in the second last chapter the basic cases of hypotheses testing are dealt with. In the last chapter a few "difficult" concepts, for example 2×2 contingency tables and elementary analysis of variance, are introduced. A last point of criticism is that there are no problems or statistical tables included for those students who prefer an "all in one" text book.

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Contiguity of Probability Measures: Some Applications in Statistics

by George G. Roussas

First edition, xiii + 248 pages. Cambridge Tracts in Mathematics and Mathematical Physics, No. 63. Cambridge University Press, 1972. U.K. £6.20

This tract presents an elaboration of the concept of "contiguity", introduced by Lucien LeCam. The concept is a powerful mathematical tool for establishing certain theoretical results with applications in statistics. The basic properties, including

asymptotic expansions and distributions of likelihood functions, are derived from the definition in Chapter 1 and these properties are repeatedly used throughout the monograph to obtain the main results. We now proceed to give a précis of each chapter.

Contiguity is a criterion of nearness of sequences of probability measures. Let $\{(X, \mathcal{A}_n)\}$ be a sequence of measurable spaces and let P_n, P'_n be probability measures on $\mathcal{A}_n$. Also, let $\{T_n\}$ be a sequence of random variables such that T_n is $\mathcal{A}_n$ -measurable. The sequences of measures P_n and P'_n are said to be *contiguous* if the following is true: for any $\mathcal{A}_n$ -measurable r.v. T_n , $T_n \rightarrow 0$ in P_n -measure if and only if $T_n \rightarrow 0$ in P'_n -measure. Some further insight into the meaning of contiguity is obtained by considering the following result: if $A_n \in \mathcal{A}_n$ then $P_n(A_n) \rightarrow 0$ if and only if $P'_n(A_n) \rightarrow 0$, provided that $\{P_n\}$ and $\{P'_n\}$ are contiguous. Some alternative characterisations of contiguity are also studied.

In the second chapter are derived some fundamental results regarding the asymptotic expansion, in the probability sense, and also the asymptotic distribution of certain likelihood functions. The underlying probability model involved is that of a Markov process satisfying certain reasonable regularity conditions. This model includes that consisting of independent identically distributed random variables. A new element introduced here is the assumption of differentiability in quadratic mean of the square root of the probability density function. This assumption replaces the usual one about the existence of two or three pointwise derivatives of the logarithm of the density. The asymptotic behaviour is considered of the log-likelihood function corresponding to an arbitrary but fixed parameter point θ and a "neighbouring" point. It is proved that asymptotically and in the neighbourhood of any parameter point the likelihood function behaves as if it were an exponential family. The rigorous justification of this assertion is in fact the main objective of Chapter 3. It is also shown there that for certain hypotheses testing problems (from an asymptotic power viewpoint) tests could be based on a random vector $\Delta_n(\theta_0)$, which depends on the derivatives in quadratic mean, mentioned earlier.

In chapter 4 are studied applications of the previous results to hypotheses testing problems where certain hypotheses and some alternatives are introduced. It is shown that in some of these cases the test based on $\Delta_n(\theta_0)$ is asymptotically uniformly most powerful and in other cases an AUMP unbiased test is obtained. The necessary assumptions for validity of the above statements are carefully listed.

Chapter 5 deals with the problem of asymptotic efficiency of a sequence of estimates of an unknown parameter. The efficiency used here is that introduced by Wolfowitz. A fair summary of the chapter could be made only if extensive additional notation were now introduced here. We restrict ourselves to mentioning that from the more general theorems stated in this chapter follows the classical result which establishes an inequality between variance of an estimator and Fisher's information function. In this particular context classical efficiency turns out to be a special case of W-efficiency.

The final chapter of the monograph is concerned with the problem of testing a simple hypothesis against a composite alternative, and contains three complicated main results. The additional mathematical notation required to describe these is quite extensive. However, the tests considered depend on the random vector $\Delta_n(\theta_0)$ referred to

earlier. It is then shown, among other things, that the sequence of tests considered, $\{\phi(\Delta_n)\}$, possesses certain optimal asymptotic properties. For instance, the power of the test ϕ on certain ellipsoidal surfaces is asymptotically constant, and within the class of tests whose power on these particular surfaces is asymptotically constant, the test ϕ is asymptotically most powerful.

An appendix of about 50 pages lists some of the basic theorems used in the body of the monograph.

To summarise, we may say that this is a research monograph, the results of which have not previously appeared in a unified way in the literature. The work is not a guide to practical procedures and would be mainly of interest to theoretical statisticians concerned with testing large sample hypotheses.

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Statistical Papers in Honour of George W. Snedecor

edited by T.A. Bancroft

First edition, xvii + 328 pages, The Iowa State University Press, Ames, Iowa, 50010, 1972 U.S. \$10.50.

This book, consisting of 19 papers by contributors who "include friends, former students, and former colleagues of Professor Snedecor", is a well-deserved (even if, from this distance, belated) tribute to G.W. Snedecor (and his famous text-book). Nevertheless, it may be a worry to some that if this trend continues, such commemorative volumes may in the future proliferate in the same way that the statistical literature has proliferated in the past 20 years.

It must be the reviewer's task to give some guidance whether the contents of the book make it worth purchasing, allowing for the possibility that with such collections it may be necessary to purchase the whole book just for one paper (cf. Berkeley symposiums)!

The contributions are rather difficult to classify, but apart from two papers on decision theory, most fall within the subject range covered by Snedecor's book and include the following topics: multiple regression, inference, experimental design, variance components, sample surveys. Many are essays, rather than papers.

Among these are interesting and very readable contributions from W.G. Cochran (observational studies), historical studies by M.G. Kendall and J.L. Lush, and one by O. Kempthorne (covering theories of inference, the Neyman-Pearson theory of testing hypotheses, etc.) which is also partly historical and decidedly illuminating.

Of the more formal papers some are frankly "potboilers", including one or two case-studies based on data of venerable age. If there are single papers which might cause one to buy the book or which will generate many references, my vote would go to the

papers of F. Yates (introducing the concept of sensitivity as an alternative representation of the power function of a test of significance) and W.T. Federer (on the construction of classes of experimental designs using transversals in Latin squares and Hedayat's sum composition method). Other papers of more than passing interest are J. Gurland and R.C. Dahiya's elegant development of an alternative to the χ^2 -test for goodness of fit of a continuous distribution and the demonstration by R.L. Anderson and his Kentucky school of the mean square error criterion as applied to the predictive efficiency of multiple regression models.

The book will appeal mostly to those statisticians whose affinities are for the sort of statistics which surfaces as statistical methods in *Snedecor's* book, particularly if they have any feeling for the history of the subject and for its status in the two pre-war decades. (For example, it is possible to deduce that the "Gertrude" to whom *Snedecor* dedicated his book, is probably his wife and not Gertrude M. Cox as many, including this reviewer, may have surmised.) It would also make an ideal presentation volume.

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